LOAN GROWTH MODEL

MARCH 2024

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Strategic Planning Department
25 MARCH 2024



Agenda

1. Introduction
2. Internal and External Factors
3. Correlation Analysis
4. Multivariate Analysis
5. Projection
6. Conclusion

Introduction

EXIM Bank's Yearly Performance

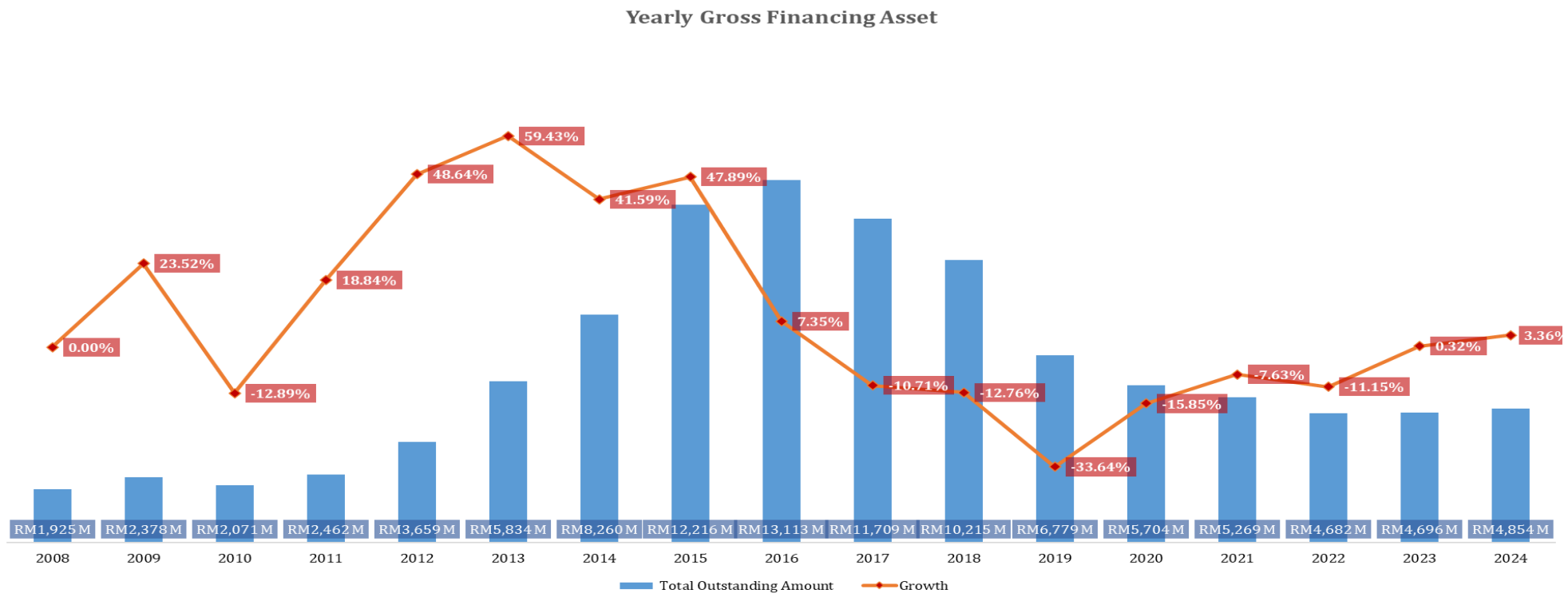
The figure shows the EXIM bank borrowing asset performance, while below shows the Measures of Central Tendency of the Yearly Gross Financing Asset Performance:

- **CAGR** Gross financing asset of **5.59%**
- **Minimum** total borrowing of **RM 1,925 M** & **-33.64%** growth
- **Maximum** total borrowing of **RM 13,113 M** & **59.43%** growth
- **Average** total borrowing of **RM 6,225 M** Quarterly & **9.14%** growth
- **Standard deviation** total borrowing of **RM 3,649 M** & **27.77**% growth

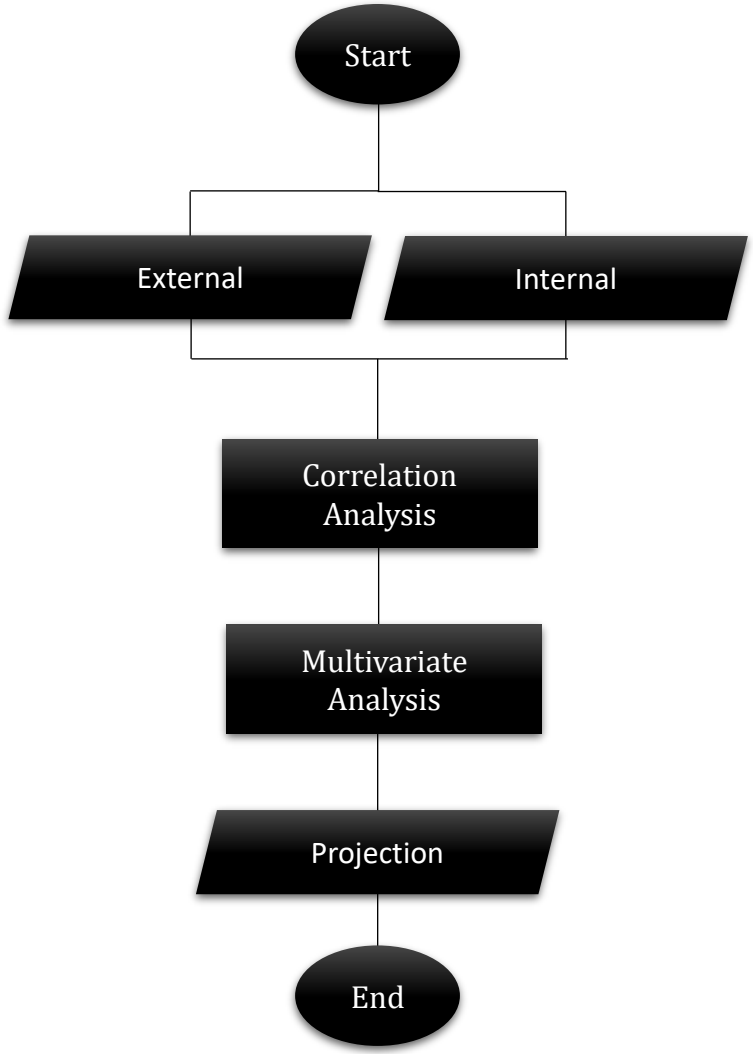
Research Questions

Based on the performance below, the research question is as follows:

- What are the main factors that driven loan growth in EXIM bank?



Process Flow



Internal and External Factors

Factors – Definition & Justification

USD Dollar Index

- Measures the value of the US dollar against a basket of six major currencies: the euro (EUR), Japanese yen (JPY), British pound (GBP), Canadian dollar (CAD), Swedish krona (SEK), and Swiss franc (CHF).

GDP (MY & US)

- Measures the total market value of all final goods and services produced within its border during a specific period.

Disbursement Status

- Measures the progress of disbursed loan in EXIM Bank

FX (USD to MYR)

- The exchange rate between the US Dollar and Malaysian Ringgit, impacting trade, investment, and economic stability in Malaysia

Fed Funds Target

- The interest rate for US Banks lending cash set by the Federal Reserve that influences the entire economy.

Nature of Accounts

- The business nature of EXIM bank's customer whether by trade and non-trade facilities

MY 10 Year Gov Bond Yield

- This is the yield for the benchmark bond maturing in November 2033, according to Bank Negara Malaysia.

FTSE Bursa Malaysia KLCI

- The benchmark stock index for Malaysia.

Brent Oil Price

- The leading global price benchmark for Atlantic basin crude oils.

USD Dollar Index: Weaking USD makes borrowing USD cheaper, potentially loan growth by lower interest rates

FX: Appreciation of RM reduces of foreign currency-denominated debt, potentially allowing for loan growth

10 Year Gov Bond Yield: Falling Yield indicates lower borrowing costs, potentially encouraging new borrowing and leading to faster loan growth

GDP: Stronger GDP indicates a growing economy with potentially higher demand for financing, leading to faster loan growth

Fed Funds Target Rate: Falling rate reduces borrowing costs and potentially encourages new borrowing, leading to faster growth

FTSE Bursa Malaysia KLCI: Rising stock market shows investor confidence and potentially easier access to capital, leading to faster loan growth

Brent Oil Price: Rising oil price for oil-exporting countries, can generate increased government revenue and potentially lead to higher loan for infrastructure projects or other investments.

Nature of Accounts: Generic segmentation of EXIM business nature

Disbursement Status: EXIM Bank's loan application process measurement up to disbursement

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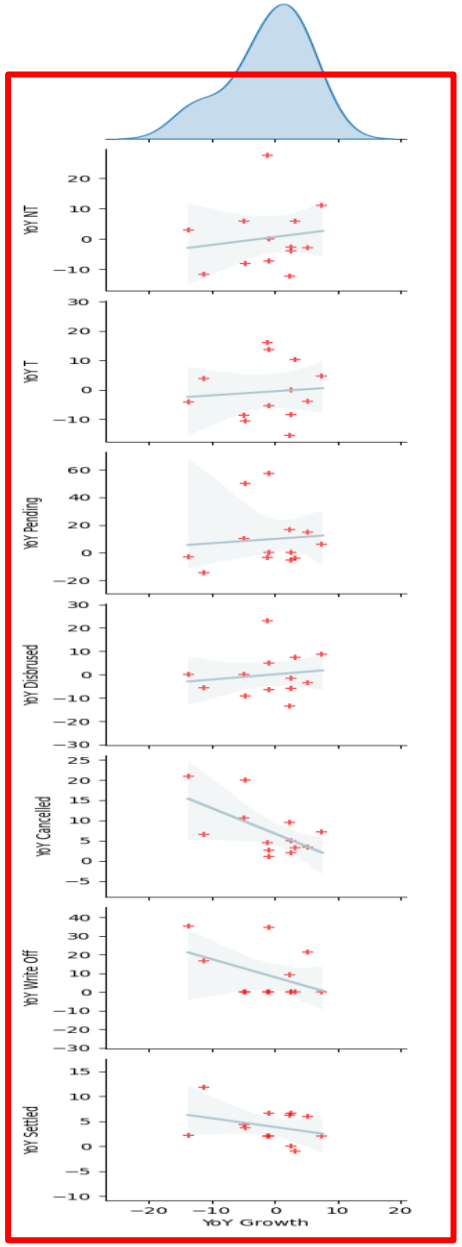
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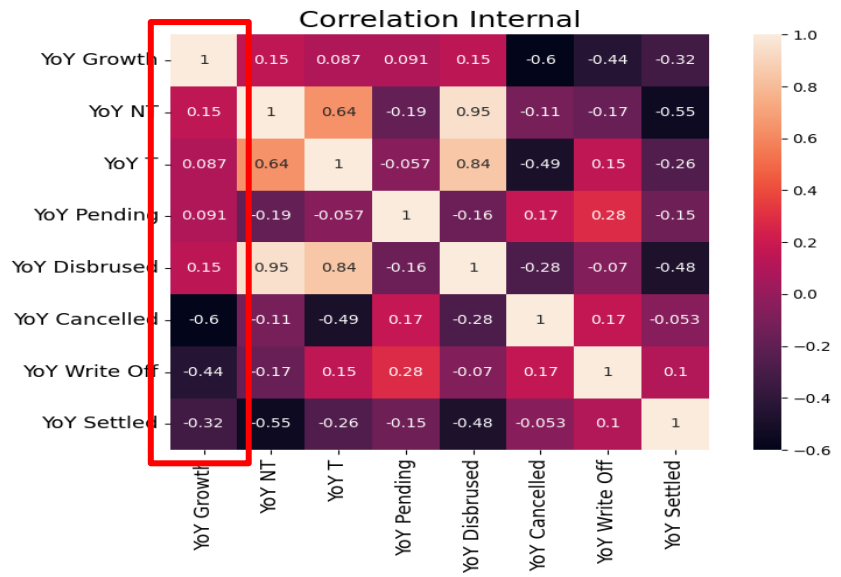
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Correlation Analysis - Internal

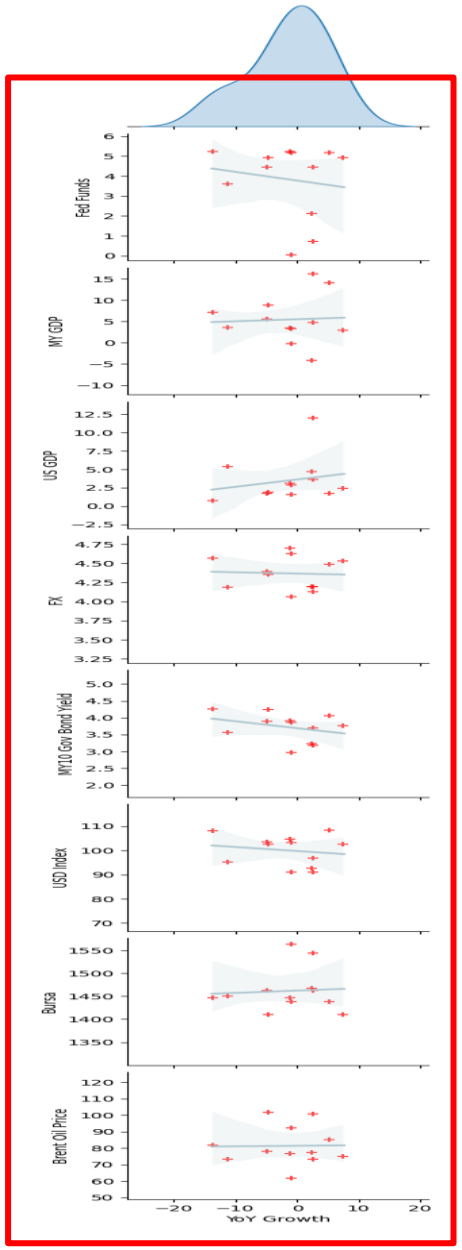


Interpretations

- The correlation shows the pending and disbursed accounts has positive contribution to the increase of gross financing asset so increase in pending and disbursed accounts will lead to increase in growth.
- By nature, non-trade has more impact with large number of sales compared to trade and it is proven by 0.15 bps and 0.09 bps respectively
- The cancelled, write off and settled accounts shows a negative correlation to the increase of gross financing asset so decrease in cancelled, write off and settled accounts will lead to increase in growth.
- To slow the settled accounts, to prevent cancellation and write-off accounts is advisable to contribute to the increase of gross financing asset.
- Most effective way to increase gross financing asset is to decrease write off accounts as the highest score of 0.44 bps.
- It is proven, EXIM bank's turn around time (TAT) upon application up to disbursement is effective when number of disbursed is 0.15 more that the pending accounts by 0.09 bps. Meaning we disbursed loan faster that we can approve application.

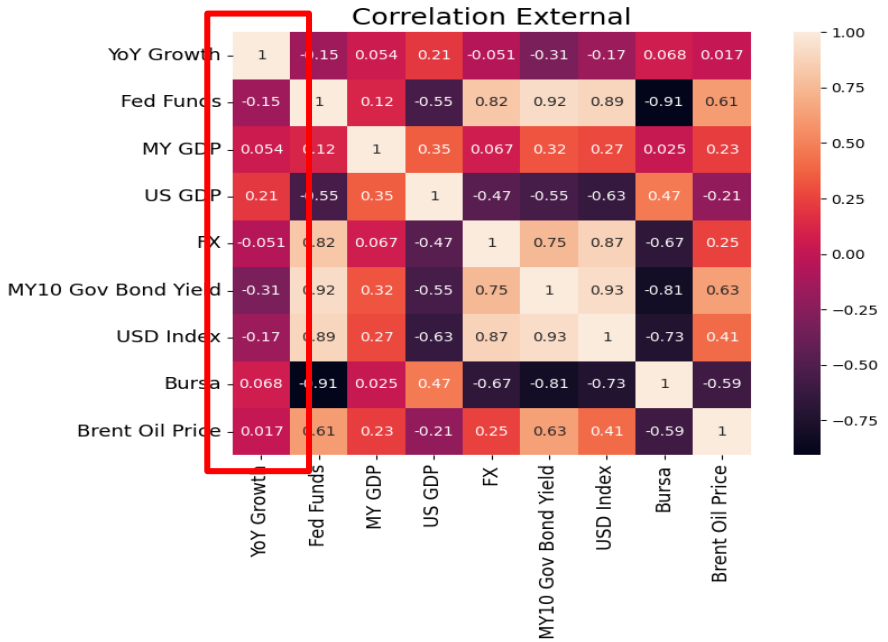


Correlation Analysis - External



Interpretations

- The correlation shows the MY GDP, US GDP and Bursa Stock Market has positive contribution to the increase of financing asset.
- The increase of US GDP is most favorable to increase to grow in asset compared to MY GDP, FX and Bursa Stock Market.
- It is predicted by most forecasts that US GDP will grow but slow in 2024 due to effects of monetary policy, larger fiscal boost, dropped in housing sector activity, supply chain bottlenecks and geopolitical risk.
- The Fed Funds Target, MY10 Gov Bond Yield and USD Index shows a negative correlation to the increase of gross financing asset.
- Most favorable scenario is for the MY10 Gov Bond Yield to decrease for the loan to grow but most forecast identify volatility in the gov bond market with an upward yield bias in 2024 due to lingering inflation and global economic volatility.



Multivariate Analysis - Internal

Projections

Growth Model on Internal Factor:

- Results shows all variable are competent do decide the direction of whether the financing asset to hike/cut by only 1 run.
- Model shows 93% accuracy
- It predicted for us to grow 10% of our financing asset is to meet conditions below:

Variable	Growth Condition
Not Trade Business	+5.30%
Trade Business	+9.45%
Pending Accounts	+72.24%
Disbursed Accounts	+7.85%
Cancelled Accounts	-10.96%
Write Off	0%
Settled Accounts	8.78%

OLS Regression Results

Dep. Variable:	YoY Growth	R-squared:	0.926
Model:	OLS	Adj. R-squared:	0.823
Method:	Least Squares	F-statistic:	8.948
Date:	Wed, 14 Feb 2024	Prob (F-statistic):	0.0141
Time:	17:07:11	Log-Likelihood:	-24.686
No. Observations:	13	AIC:	65.37
Df Residuals:	5	BIC:	69.89
Df Model:	7		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	9.2498	1.868	4.952	0.004	4.448	14.052
YoY NT	-18.6474	5.033	-3.705	0.014	-31.584	-5.710
YoY T	-12.1191	3.171	-3.822	0.012	-20.271	-3.968
YoY Pending	0.1422	0.043	3.273	0.022	0.030	0.254
YoY Disbrused	30.5292	8.187	3.729	0.014	9.484	51.574
YoY Cancelled	-0.7637	0.157	-4.850	0.005	-1.169	-0.359
YoY Write Off	0.0845	0.085	0.999	0.364	-0.133	0.302
YoY Settled	-1.1064	0.315	-3.511	0.017	-1.916	-0.296

Omnibus:	1.157	Durbin-Watson:	1.645
Prob(Omnibus):	0.561	Jarque-Bera (JB):	0.741

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Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Multivariate Analysis - External

Projections

Growth Model on External Factor:

- Results shows variable are competent with little adjustment do decide the direction of whether the financing asset to hike/cut by 2 run.
- Model shows 93% accuracy
- It predicted for us to grow 10% of our financing asset is to meet conditions below:

Variable	Growth Condition
Fed Funds	2.68
GDP (MY)	-12.04
GDP (US)	14.68 in DF2
FX	0.19 in DF
MY10 Gov Bond Yield	-0.61 in DF2
USD Rates	-4.57 in DF
Bursa Stock	1323.36
Brent Oil Price	108.0

OLS Regression Results

Dep. Variable:	YoY Growth	R-squared:	0.926
Model:	OLS	Adj. R-squared:	0.730
Method:	Least Squares	F-statistic:	4.711
Date:	Mon, 19 Feb 2024	Prob (F-statistic):	0.115
Time:	16:06:10	Log-Likelihood:	-22.971
No. Observations:	12	AIC:	63.94
Df Residuals:	3	BIC:	68.31
Df Model:	8		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	156.1288	97.795	1.596	0.209	-155.098	467.355
Fed Funds	-5.6740	1.720	-3.299	0.046	-11.148	-0.200
MY GDP	-0.9428	0.398	-2.367	0.099	-2.210	0.325
US GDP (DF2)	0.9725	0.352	2.762	0.070	-0.148	2.093
FX (DF)	124.6216	24.870	5.011	0.015	45.473	203.770
MY10 Gov Bond Yield (DF2)	-26.5592	5.070	-5.238	0.014	-42.695	-10.424
USD Index (DF)	-2.4927	0.653	-3.820	0.032	-4.570	-0.416
Bursa	-0.1177	0.064	-1.848	0.162	-0.320	0.085
Brent Oil Price	0.4639	0.155	2.997	0.058	-0.029	0.957

Omnibus:	1.078	Durbin-Watson:	2.159

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Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 1.51e+05. This might indicate that there are strong multicollinearity or other numerical problems.

Conclusion

Statement

• Main factors that driven the loan to increase is the strengthen of US GDP, but it is unlikely for US to strengthen rapidly rather it will grow slowly due to effects of monetary policy, larger fiscal boost, dropped in housing sector activity, supply chain bottlenecks and geopolitical risk.

• Main factors that prevent the loan to increase is the increase of write off accounts with CAGR 6.72%. Several accounts latest in 2023 are Alami Vegetable Oil Products Sdn Bhd, Fresh Bowls Limited, Hartamanis (S) Pte Ltd, I Leaves (UK) Ltd, Malaysian Delicacies Pty Ltd, Gold Rain Resources Sdn Bhd, Quantum Orient Holdings Sdn Bhd and Penang Civil Storage Company Sdn Bhd.

• By nature, non-trade has more impact compared to trade due to large amount of sales.

• It is proven, EXIM bank's turnaround time (TAT) upon application up to disbursement is improving when number of disbursed is 0.15 more that the pending accounts by 0.09 bps. Meaning we disbursed loan faster that we can approve application.

Thank You